Sawan Kumar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.61
A.B.S INTER COLLEGE,LALGANJ,VAISHALI	Class XII (BSEB)	2017	65.40%
R K HIGH SCHOOL KHUTAHIN,BHATAULIA	Class X (BSEB)	2015	65.80%

PROJECTS

Master's Thesis: Al-Powered Real-Time Vein Visualization Application | Python, OpenCV, PyTorch, YOLO, Raspberry Pi ,Android Studio, Edge Al

- **Computer Vision Pipeline**: Engineered a real-time video processing pipeline in Python (OpenCV) using CLAHE onntrast enhancement and adaptive thresholding, reducing image noise by 40% across 1,000+ benchmarked frame matrices.
- **Model Optimization & Accuracy**: Trained and evaluated lightweight U-Net and YOLO segmentation models in PyTorch, achieving 92.4% mAP and an lot of 0.86 for subcutaneous vein localization under low-light conditions.
- **Edge Hardware Deployment**: Successfully deployed and benchmarked the deep learning pipeline on Raspberry Pi, achieving a stable processing rate of 30 FPS and maintaining edge latency under 33 ms per frame.
- **Ongoing Mobile App Development**: Currently developing a dedicated Android application to interface directly with external USB/micro-cameras via OTG, targeting real-time augmented contrast overlays to reduce clinical vein location search time by 25%.

Ophthalmic Patient Risk Stratification & Diagnostic Analytics | Python ,Pandas, Scikit-Learn, SQL, Power Bi/Tableau Descriptive Statistics(LVPEI)

- **Data Pipeline & Cleaning**: Designed SQL queries and data validation workflows to clean, aggregate, and structure 15.000+ unstructured clinical records, reducing data preprocessing latency by 35%
- **Predictive Modeling & Accuracy**: Developed classification models (Random Forest. Logistic Regression) on ocular metrics (IOP, corneai thickness) to detect early-stage glaucoma, achieving 88% precision and an **AUC-ROC of 0.91.**
- **Business & Clinical Impact**: Built interactive dashboards to visualize patient risk tiers and screening throughput, accelerating diagnostic review time by **25%** for clinical teams.

TRAINING & EXPERIENCE

Quality Assurance & Engineering Intern | Sudha Dairy

- **Quality Control & Chemical Testing:** Conducted routine chemical composition testing (fat %, SNF, density) on milk batches and logged quality results.
- **Data Analysis & Quality Reporting:** Compiled daily test data in MS Excel to track batch variance trends and created weekly quality summary reports for plant management.
- **Engineering Maintenance (Boiler & Freezing):** Monitored operational parameters, temperature logs, and routine maintenance for plant boilers and refrigeration/freezing systems.
- **Packaging Line Operations:** Supervised packaging line equipment, tracking machine downtime and daily output metrics to ensure smooth plant operations.

SKILLS

Data Analysis & Machine Learning: Python (Pandas, NumPy, Scikit-Learn, PyTorch), OpenCV, SQL, Data Visualization.

Engineering Analytics: Process Data Modeling, Time-Series Sensor Analytics, Quality Control Metrics

Tools & Frameworks: Git, MS Excel, Tableau, ArcGIS, MATLAB, Zemax, CATIA V5

POSITIONS OF RESPONSIBILITY

Head Placement Coordinator | Centre for Interdisciplinary Program (CIP), IIT Hyderabad

- Leading placement and internship drives across all **7 branches** under CIP, serving as the primary liaison between corporate recruiters, department faculty, and student candidates.
- Managing recruiter outreach, coordinating candidate shortlisting, and overseeing seamless portal operations for multi-disciplinary batches.

Graduate Teaching Assistant | IIT Hyderabad

- Assisting course faculty in assignment evaluation, lab session management, and student query resolution for **OP5033: Applied Optics**.

RELEVANT COURSES

AI in Ophthalmic Informatics, Machine Learning and Its Applications, Computational and Numerical Methods,

Applied Optics and Optical System Design, Laser Physics and Applications, Optoelectronics and Photonics,

Biomedical Imaging and Image Processing, Optical Design and Simulation Using Zemax